

Flow and Flux

Line integrals over vector fields have the natural interpretation of computing work when $\vec{F}$ represents a force field. It is also common to use vector fields to represent velocities.

The line integral $\int_C \vec{V} \cdot d\vec{s}$ is said to represent the **flow**.

When C is a closed curve, we call flow **circulation**, represented by

$$\oint_C \vec{V} \cdot d\vec{s}$$

The “opposite” of flow is **flux**, a measure of “how much water is moving *across* path C .” If a curve represents a filter in flowing water, flux measures how much water will pass through the filter. The flux is represented as,

$$\int_C \vec{V} \cdot \vec{n} \, ds$$

Where $\vec{n}$ is a unit vector orthogonal to the curve C .

Green’s Theorem

There is an important connection between the circulation around a closed region R and the curl of the vector field inside of R , as well as a connection between the flux across the boundary of R and the divergence of the field inside.

Green’s Theorem states “the counterclockwise circulation around a closed region R is equal to the sum of the curls over R .”

$$\oint_C \vec{V} \cdot d\vec{s} = \iint_R \text{curl } \vec{V} \, dA$$

Stokes Theorem

The Stoke's theorem states that “the surface integral of the curl of a function over a surface bounded by a closed surface is equal to the line integral of the particular vector function around that surface.”

$$\oint_C \vec{V} \cdot d\vec{s} = \iint_S (\text{curl } \vec{V}) \cdot \vec{n} \, dS$$